

# OLUWAMAYOWA SAMUEL ADEBAMBO, MIET, MNSE

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## Profile

Robotics and autonomous-systems engineer with experience in real-time perception, embedded vehicle control, sensor fusion and EV powertrain development. **Perception Lead for the Formula Student AI UK 2026 Overall Championship-winning team**, contributing across perception, localisation and mapping, VCU software and powertrain integration.

## Technical Skills

**Programming:** Python, C++, Embedded C, SQL

**Robotics/AI:** ROS 2, ORB-SLAM3, OpenCV, YOLO26, ResNet-50, PaddleOCR, TensorRT, PyTorch, SmolVLA, faster-whisper, LiDAR, stereo vision, EKF/CKF, sensor fusion

**Embedded/Engineering:** ESP-IDF, FreeRTOS, CAN bus, NVIDIA Jetson, Docker, Git, SolidWorks, MATLAB

## Work Experience

**Oxford Brookes Racing Autonomous** | *Perception Lead; VCU & Powertrain Engineer*

*Sep 2025 – Present*

**Formula Student AI UK 2026 – Overall Winner**

Oxford, UK

- Led the autonomous race-car perception pipeline, deploying YOLO-based cone detection on NVIDIA Jetson and reducing inference latency from 284 ms/frame to 20 ms FP16 and 14 ms INT8 (93–95% reduction).
- Contributed to the localisation and mapping pipeline by implementing CKF fusion of wheel odometry and LiDAR pose estimates; developed monocular cone localisation and evaluated camera–LiDAR fusion using regression and least-squares methods.
- Developed and debugged ESP-IDF/FreeRTOS VCU software in embedded C, including CAN messaging, handshake logic, autonomous state transitions, ASSI indicators, buzzer timing and emergency-shutdown behaviour.
- Contributed to EV powertrain design and integration through SolidWorks drivetrain components, reduction-system concepts and simulation-led analysis.

**Oxford Brookes ADIT Research Group** | *Research Assistant*

*Jan 2026 – Present*

Oxford, UK

- Investigate sensor-fusion and visual-inertial localisation methods for autonomous systems, focusing on robust perception and localisation in real-world robotic environments.
- Configure and deploy perception pipelines on NVIDIA Jetson Orin Nano, managing Ultralytics and dusty-nv Docker containers for repeatable embedded-inference workflows.
- Assembled a Hiwonder MentorPi A1 robotic platform, integrating sensors, compute and control systems; supported calibration, testing and validation of perception and control modules.

**Payvance Ltd** | *Data Science Intern*

*Oct 2024 – Aug 2025*

Nigeria

- Built end-to-end Python and SQL pipelines to collect, clean and prepare transaction data at scale for analysis and reporting.
- Developed management dashboards in Power BI and Python using Matplotlib and Seaborn, replacing manual reporting processes with repeatable analytical workflows.
- Standardised data-cleaning procedures and performed exploratory analysis to identify trends, anomalies and operational insights for business decision-making.

**Roadmechanisms Ltd** | *Automobile Engineering Lead Intern*

*Feb 2023 – Sep 2023*

Nigeria

- Led a team of five engineering interns across daily vehicle diagnostics, repairs and preventive-maintenance scheduling, coordinating workshop tasks and technical activities.
- Conducted mechanical and electrical fault diagnosis using equipment including the Launch X-431 IV and a computerised wheel balancer; communicated findings and repair requirements to customers.

## Education

**Oxford Brookes University** – MSc Data Science and Artificial Intelligence

*Expected Sep 2026*

**Federal University of Agriculture Abeokuta** – BEng Mechanical Engineering

*Sep 2024*

## Projects

**HumanSLAM – Robust Localisation under Perceptual Aliasing and Divergence** (*MSc Dissertation*)

Python · C++ · ROS 2 · ORB-SLAM3 · ResNet-50 · YOLO26 · PaddleOCR · ZED 2i · Oxts IMU

- Designing a cascaded three-gate confidence architecture over ORB-SLAM3, combining geometric, semantic and linguistic evidence to reduce false loop closures and localisation divergence in perceptually repetitive environments.
- Integrating ResNet-50 scene classification, YOLO26 segmentation for dynamic-object filtering and spatial matching, and PaddleOCR text anchors within a ROS 2 localisation pipeline.
- Benchmarking HumanSLAM against ORB-SLAM3 on Oxford RobotCar and KITTI using ATE, RPE and loop-closure robustness measures.

**Voice-Controlled Vision-Language-Action Off-Road Rover** (*In Progress*)

Python · ROS 2 · SmolVLA · faster-whisper · Computer Vision · Wheel Odometry

- Developing an embodied-AI control system by preparing demonstrations containing camera observations, natural-language instructions, linear/angular velocity state and corresponding rover actions for SmolVLA fine-tuning.
- Designing a ROS 2 pipeline for synchronised episode collection and integrating faster-whisper for local speech-to-text transcription, converting spoken commands into language prompts for voice-controlled operation.

**Autonomous Solar Lighting System – Embedded Control**

Embedded C · Arduino · LDR Sensor · Photovoltaic Systems · Relay Control

- Designed and built an off-grid solar lighting system with autonomous dusk-to-dawn operation, integrating photovoltaic power management with real-time LDR sensing and relay actuation in embedded C.

## Professional Memberships

**IET Member Research Group (MRG)**

*Selected Member, 2026 – Present*

- Selected as one of 58 members to join the IET Member Research Group.
- Contribute to research activities on engineering and technology challenges, collaborating with engineers, researchers and industry professionals to inform IET initiatives and represent early-career perspectives.

**Member (MIET), Institution of Engineering and Technology | Member (MNSE), Nigerian Society of Engineers**